Fiscal Year:	FY 2009	**Task Last Updated:**	FY 03/31/2009
PI Name:	Pluth, Janice Ph.D.		
Project Title:	Fundamental Biological Studies Of Protein Phosphorylation Profiles After HZE Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	janice.pluth@unlv.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	702-895-3236
Organization Name:	University of Nevada, Las Vegas		
PI Address 1:	Department of Health Physics and Diagnostic Sciences		
PI Address 2:	4505 S. Maryland Pkwy, Mail Code: 3037		
PI Web Page:			
City:	Las Vegas	**State:**	NV
Zip Code:	89154-9900	**Congressional District:**	1
Comments:	NOTE: Formerly at Lawrence Berkeley National Laboratory until fall 2017.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	09/01/2004	**End Date:**	12/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA ARC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 4/2009) NOTE: Received NCE to 12/31/2008 per A. Chu/ARC (8/08) NOTE: added grant #NNA06CD66I as an additional grant number, per A. Chu/ARC (6/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNA04CL69I ; NNA06CD66I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA is interested in understanding the biological consequences of exposure to constituents of space radiation, such as protons and HZE particles, and identifying possible means of intervention. Biological processes of interest include DNA damage processing, signal transduction, and cell cycle control. Despite state-of-the-art physical dosimetry capabilities, accurately measuring energy deposition cannot, and does not, account for different individual genetic backgrounds, specific tissue sensitivities, or bystander effects. All of these factors are critical variables of the highest significance for accurate risk assessment. We propose to establish in vivo molecular response profiles to HZE exposure and bystander effects, whereby these complex variables will inherently be accounted for by measuring the resulting cellular manifestations. Within minutes of ionizing radiation exposure, a number of phosphorylation events occur, orchestrating cellular signaling pathways critical for cell survival, homeostasis, and genomic stability, thus determining the ensuing fate of cells after damage. We will detail phosphorylation events after Fe ion exposure to establish a molecular profile of the biological response to this particularly relevant HZE particle. Standard immunofluorescence microscopy and a novel fluorescence activated cell sorter (FACS) based assay will be used to quantitiate phosphorylation levels. The FACS based assay is a non-subjective, relatively rapid method capable of quantitating phosphorylation levels in specific phases of the cell cycle, making it ideal for these studies. Profiles of molecular markers, phosphorylated in response to Fe ion and X-ray exposure will be compared at various times to DNA repair, cell cycle progression and ultimately survival. Bystander effects, which are effects occurring in cells not directly exposed to radiation, could account for a substantial part of the total risk associated with space irradiation. Therefore, it is essential to define molecular phosphorylation profiles in response to high and low LET induced bystander effects for accurate risk assessment. These profiles will be compared to each other and to direct exposure profiles to determine unique characteristics associated with these different stimuli, and to establish quantifiable endpoints for the bystander effect. To identify potential countermeasures, we will also test whether addition of anti-oxidants diminishes the observed direct radiation and bystander effects. Different tissues of the body are known to respond differently to ionizing radiation; the effects of HZE in this regard are not well understood. We will use two tissue representative cell types, fibroblasts and lymphocytes, to determine the universality of the molecular phosphorylation profiles. Lastly, we will take advantage of rapidly advancing studies to develop novel phospho-specific antibodies to a lymphocyte specific protein. In total, this work will provide quantitative profiles of the biological response and establish HZE-induced molecular fingerprints. Establishing this type of bio-dosimetry holds promise for development of a blood-based assay, which could be invaluable in accurately assessing an individuals' risk associated with space travel. Ultimately, implementation of this knowledge may lead to accurate bio-dosimetry, quantitation of the bystander effect, and development of countermeasures, all of which can be applied to evaluate and reduce the risks associated with space radiation exposures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
	Brief description of scientific goals/objectives of the proposed research. • The experiments performed in this project were designed to test the hypothesis that that initiation and the time course of phosphorylation signals, survival, and checkpoint responses differ between low and high-LET radiation. We also hypothesized that bystander effects, effects postulated to occur in cells not directly exposed to radiation, but exposed to deleterious factors they secrete, may cause specific signaling events leading to unique phosphorylation profiles. Lastly, we hypothesized that cells from different tissue origins may elicit differences in dose-response or temporal patterns of phosphorylation. • Normal hTERT fibroblast and lymphoblast cells were exposed to low and high LET radiation and detailed phospho-protein profiles were obtained. Both dose and time course response for three phospho-proteins providing optimal specificity and signal to background noise were determined. These experiments were designed to allow us to determine whether cell signaling following high and low LET exposures is distinct. The disparity in the kinetics observed between these radiation qualities may ultimately lead to a better understanding of how astronaut cells' may respond differentially to these exposures, providing a better understanding of the risk of high LET exposures. • The project also involved testing for bystander effects. It has been suggested that bystander effects will dominate at low doses and dose rates at which astronauts are exposed, since only a small proportion of all cells will experience a hit from an HZE particle, and the remaining cells will be bystanders. In this project we compared bystander responses using a co-culture system and AG1522 normal fibroblast cells, previously published to show a bystander effect. These studies were initiated to ideally discover unique phospho-protein profiles specific to bystander exposures. • Lastly, a fairly newly discovered protein, Artemis, identified as being important in resolving radiation damage was investigated in these studies. We proposed producing phospho-specific antibodies for this protein to then use to define the phospho-protein profiles following low and high LET exposures. Some evidence for Artemis being particularly important in resolving complex damages, observed more frequently following high LET exposures has been noted. Therefore these studies could provide for a unique biomarker to study high LET exposures in lymphocytes (Artemis is expressed at low levels in most tissues with the exception of lymphocytes). This report includes results from the NSRL-5, NSRL-7, NSRL06A, NSRL06B, NSRL07A, NSRL07B, NSRL08A, and NSRL08C. Brief summary of project's research accomplishments. Aim 1. Flow cytometry assays were optimized for 3 different phospho-proteins (gamma-H2AX Ser139, pATF2 Ser 490/498 and pSMC1 Ser 957). Fibroblasts were irradiated with a range of doses (0.05, 0.1, 0.5 and 2 Gy) using Fe (1000 MeV/u) at BNL and Xray (320 keV) at LBNL. Data was obtained for three independent Xray and Fe ion experiments and analyzed by flow cytometry. We have observed that fibroblasts show reduced ATM-mediated phosphorylations following high-LET as compared to low-LET, and have recently published this work (Radiat Res. 2008 Sep;170(3):353-64). Subsequent studies were initiated to determine whether all ATM mediated signaling was reduced following high LET exposures using foci studies for antibodies that could not be optimized for flow cytometry staining. Preliminary results confirm our previous FAC results with pATF2 and pSMC1, revealing 53BP1 phosphorylation to also be reduced following high LET as compared to low LET exposures. Data from flow cytometry experiments have also been used in collaborative studies with Frank Cucinotta and others at Johnson Space Center. Computer based methods have been optimized to identify cells in particular phases of

Task Progress:	the cell cycle and to statistical analyze these large sets of individual cell data. Data resulting from these collaborative studies is currently being compiled for a paper in preparation. Analysis of individual cell data has also been very useful in identifying differences between exposed and non-exposed cells even at very low doses (0.05 Gy). Aim 2: For bystander experiments a co-culture method and AG1522 normal fibroblasts were used. Flow cytometry was used to determine levels of p21, gamma-H2AX and ROS in control, direct-exposed and bystander cells. One Xray, three Fe and one Ti experiment have been performed thus far. Aim 3. Flow-based methods of staining was optimized for lymphoblasts for each of the antibodies used in fibroblasts studies. Three independent Xray and Fe exposures were completed. Distinct differences in induction and kinetics of phosphorylation between fibroblasts and lymphoblasts were noted for gH2AX. Following Fe exposure lymphoblasts show a greater induction and degree of residual damage following irradiation as compared to fibroblasts. However following Xray exposure lymphoblasts show a lower induction per dose as compared to fibroblasts. Both cell types show a greater induction of gH2AX following Fe as opposed to Xray. Fibroblasts studies had shown a greater induction of ATM mediated phosphorylations following low LET exposures but lymphoblasts cells showed a very high degree of phosphorylation even without any damage, thus very minor levels of induction were observed following either Fe or Xray exposure. Phospho-antibodies studies of exposed and non-exposed cells lymphocytes visualized on microscope slides also revealed a large number of foci in non-exposed cells for pATF2 and pSMC1. These results were unexpected and it is uncertain why unexposed cells would have pATF2 and pSMC1 foci that have been published to be specific to damaged cells. We are currently investigating this finding further with Western blotting studies. We have performed both BrdU survival studies and observed phospho-specific kinetics of gH2AX in wild type and over-expressing Artemis cells. These studies have revealed that Artemis over-expression decreases radiation sensitivity and that gH2AX levels are decreased slightly more rapidly at earlier times post damage. Discussion: Major conclusions that can be drawn from this work are that high and low LET exposures initiate different signaling. In particular, ATM mediated signaling is reduced following high LET exposure as compared to low LET. A failure to fully activate ATM repair pathways may lead to a delay or absence of repair that could lead to mis-repair and cell death respectively. The critical road map asks what are the probabilities for increased carcinogenesis from space radiation as a function tissue and radiation quality? This work would suggest that there is an increased risk of carcinogenesis following high LET given that proper signaling of damage does not occur following this type of exposure. Our studies also suggest that different tissues respond differently to these exposures, with lymphoblasts showing more residual damages at 24 h post Fe ion exposure. Flow based studies comparing lymphoblast and fibroblast phospho-kinetic responses to low and high LET are relevant to the critical road map question of “how can tissue specific probabilities for increased carcinogenesis risk from space radiation be best evaluated?” From these studies it seems that certain signaling pathways are not specific for damage in all cell types, as noted by the pATF2 and pSMC1 foci observed in control lymphoblast cells. Whether the lack of pATF2 and pSMC1 induction following radiation damage in lymphocytes results in differences in carcinogenic potential for these cells is uncertain and would require additional studies. From these studies we can only confirm that different cell types do not respond identically to these radiation exposures. Our bystander experiment results point us to a partial response to the critical road map question of “whether bystander effects modulate the expression of radiation damage and how can the understanding of mechanisms be used to predict carcinogenic risks from space radiation?” We have confirmed that gamma-H2AX is induced in bystander exposed cells, inducing roughly 1.5 fold increase in gamma-H2AX over control levels following co-culture with cells exposed to 2.5 Gy Fe ion. DSBs induced in bystander cells may be especially important at low doses, such as those received by astronauts, as more of their cells will be receiving bystander exposures as compared to direct exposures. Time course studies would be needed to determine whether bystander induced breaks are repaired with similar kinetics as direct track damage, and this information may be important in understanding the carcinogenic risks for these exposures. In subsequent studies, we plan to follow up on this work in the near future to determine whether ATM mediated signaling in bystander-exposed cells is different from directly exposed cells. These studies also confirm Artemis’ importance in resolving radiation damage and reveal that over-expression of this protein can lead to a reduction in radiation sensitivity, both following high and low LET exposures. It is uncertain whether polymorphisms exist in this gene resulting in an increased or decreased expression of this protein, but if they do exist it would definitely affect an individual’s sensitivity to radiation and likely to radiation induced carcinogenesis as well. In studies for our recently funded NASA grant we have extended our flow cytometry studies to determine whether there are unique phospho-profiles based on LET, and to how clustered damage and direct track vs delta ray damage may differentially effect cell signaling. Preliminary data appear to show that more complex damages, or increasing direct track to delta ray ratios reduces that level of ATM mediated signaling, at least at early times post exposure. These additional studies should be of great importance in better understanding cellular effects of the wide spectrum of ions and energies which astronauts are exposed to.

Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Cucinotta FA, Pluth JM, Anderson JA, Harper JV, O'Neill P. "Biochemical kinetics model of DSB repair and induction of gamma-H2AX foci by non-homologous end joining." Radiat Res. 2008 Feb;169(2):214-22. PMID: 18220463 , Feb-2008
Articles in Peer-reviewed Journals	Whalen MK, Gurai SK, Zahed-Kargaran H, Pluth JM. "Specific ATM-mediated phosphorylation dependent on radiation quality." Radiat Res. 2008 Sep;170(3):353-64. PMID: 18763865 , Sep-2008
Articles in Peer-reviewed Journals	Pluth JM, Yamazaki V, Cooper BA, Rydberg BE, Kirchgessner CU, Cooper PK. "DNA double-strand break and chromosomal rejoining defects with misrejoining in Nijmegen breakage syndrome cells." DNA Repair (Amst). 2008 Jan 1;7(1):108-18. PMID: 17919995 , Jan-2008
Articles in Peer-reviewed Journals	Shimura T, Martin MM, Torres MJ, Gu C, Pluth JM, DeBernardi MA, McDonald JS, Aladjem MI. "DNA-PK is involved in repairing a transient surge of DNA breaks induced by deceleration of DNA replication." J Mol Biol. 2007 Mar 30;367(3):665-80. Epub 2007 Jan 12. Erratum in: J Mol Biol. 2007 Jul 27;370(5):1033. DiBernardi Maria A [corrected to DeBernardi, Maria A]. PMID: 17280685 , Mar-2007
Articles in Peer-reviewed Journals	Wang J, Pluth JM, Cooper PK, Cowan MJ, Chen DJ, Yannone SM. "Artemis deficiency confers a DNA double-strand break repair defect and Artemis phosphorylation status is altered by DNA damage and cell cycle progression." DNA Repair (Amst). 2005 May 2;4(5):556-70. PMID: 15811628 , May-2005

Articles in Peer-reviewed Journals	George KA, Hada M, Jackson LJ, Elliott T, Kawata T, Pluth JM, Cucinotta FA. "Dose response of gamma rays and iron nuclei for induction of chromosomal aberrations in normal and repair-deficient cell lines." Radiat Res. 2009 Jun;171(6):752-63. http://dx.doi.org/10.1667/RR1680.1 ; PubMed PMID: 19580482 , Jun-2009
Articles in Peer-reviewed Journals	Sridharan DM, Chappell LJ, Whalen MK, Cucinotta FA, Pluth JM. "Defining the biological effectiveness of components of high-LET track structure." Radiat Res. 2015 Jul;184(1):105-19. http://dx.doi.org/10.1667/RR13684.1 ; PMID: 26114329 , Jul-2015
Articles in Peer-reviewed Journals	Sridharan DM, Chien L-C, Cucinotta FA, Pluth JM. "Comparison of signaling profiles in the low dose range following low and high LET radiation." Life Sci Space Res (Amst). 2020 May;25:28-41. https://doi.org/10.1016/j.lssr.2020.02.002 ; PMID: 32414491 , May-2020